Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: JNM Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
7742 Paradise RD	Modesto	Stanislaus	95358

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 11/15/2013

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X017-X027-X040-XXXX

B. OPERATORS

Morris, John

Operator name: Morris, John

Telephone no.: Landline: ; Cellular: (209) 404-3909

Mailing Address Number and Street	City	State	Zip Code
7742 Paradise RD	Modesto	CA	95358

This operator is responsible for paying permit fees.

Morris, Nicole

Operator name: Morris, Nicole

Telephone no.: Landline: ; Cellular: (209) 814-5392

Mailing Address Number and Street	City	State	Zip Code
7742 Paradise RD	Modesto	CA	95358

C. OWNERS

Morris, John

Legal owner name:	Morris, John	Telephone no.:		(209) 404-3909
			Landline	Cellular
7742 Paradise RD		Modesto	CA	95358
Mailing Address Number and Street		City	State	Zip Code

This owner is responsible for paying permit fees.

Morris, Nicole

Legal owner name:	Morris, Nicole	Telephone no.:		(209) 814-5392
			Landline	Cellular
7742 Paradise RD		Modesto	CA	95358
Mailing Address Number and Street		City	State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	150	50	100
Number under roof	470	70	0	0	0	0
Maximum number	475	75	0	150	50	100
Average number	470	70	0	150	50	100
Avg live weight (lbs)	1,400	1,450	0	650		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 15,105.63 *tons per reporting period*

Total nitrogen from manure: 193,746.04 *lbs per reporting period*

After ammonia losses (30% loss applied): 135,622.23 *lbs per reporting period*

Total phosphorus from manure: 31,588.12 *lbs per reporting period*

Total potassium from manure: 90,626.92 *lbs per reporting period*

Total salt from manure: 237,396.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 3,049,500 *gallons*

Total nitrogen generated: 19,296.30 *lbs*

Total phosphorus generated: 2,177.00 *lbs*

Total potassium generated: 23,341.24 *lbs*

Total salt generated: 120,661.94 *lbs*

	3,049,500	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	3,049,500	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
MID Lateral 5	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/12/2023	Solid commercial fertilizer *4-0-0*	4.60 *ton*	Dry-weight	0.1	4.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	367.63	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	367.63	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	Corral solids	1,610.00 *ton*	Dry-weight	48.8		20,200.00	9,040.00	29,100.00		53.40

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	33,302.53	14,903.71	47,975.42	880,373.76
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	33,302.53	14,903.71	47,975.42	880,373.76

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	12	12	0	none	X017-X027-X040-XXXX
Field 2	17	17	2	process wastewater	X017-X027-X040-XXXX
Field 3	16	16	2	process wastewater	X017-X027-X040-XXXX
Field 4	7	7	2	process wastewater	X017-X027-X040-XXXX
Field 5	6	6	2	process wastewater	X017-X027-X040-XXXX
Field 6	8	8	0	none	X017-X027-X040-XXXX
Field 7	5	5	0	none	X017-X027-X040-XXXX
Totals for areas that were used for application	46	46	8		
Totals for areas that were not used for application	25	25	0		
Land application area totals	71	71	8		

B. CROPS AND HARVESTS

Field 2

Field name: Field 2

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	0.10 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	0.01	0.00	0.00	0.00	0.00

Field 2

07/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 07/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/03/2023	462.00 *ton*	As-is		71.4	3,740.00	870.00	4,050.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	27.18	203.28	47.29	220.13	1,134.78

Field 3

Field name: Field 3

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 16 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	0.10 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	0.01	0.00	0.00	0.00	0.00

07/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 07/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/03/2023	429.00 *ton*	As-is		71.4	3,740.00	870.00	4,050.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	26.81	200.56	46.65	217.18	1,119.58

Field 4

Field name: Field 4

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 7 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	0.10 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	0.01	0.00	0.00	0.00	0.00

07/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 7 Plant date: 07/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/03/2023	206.00 *ton*	As-is		75.6	3,670.00	640.00	2,980.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	29.43	216.01	37.67	175.39	775.50

Field 5

Field name: Field 5

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 6 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	0.10 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	0.02	0.00	0.00	0.00	0.00

Field 5

07/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 6 Plant date: 07/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/03/2023	167.00 *ton*	As-is		75.6	3,670.00	640.00	2,980.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	27.83	204.30	35.63	165.89	733.46

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 2 - 11/17/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough — Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	161.17	19.14	89.82	322.33	171,000.00 *gal*
Application event totals		161.17	19.14	89.82	322.33	

Field 2 - 07/12/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage — Plant date: 07/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	85.26	9.31	135.58	754.07	313,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	53.94	2,970,000.00 *gal*
Application event totals		85.26	9.31	135.58	808.01	

07/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	41.68	2,295,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.68	

Field 2 - 07/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.00	6.77	98.60	548.41	228,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	39.23	2,160,000.00 *gal*
Application event totals		62.00	6.77	98.60	587.65	

09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.00	6.77	98.60	548.41	228,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	39.23	2,160,000.00 *gal*
Application event totals		62.00	6.77	98.60	587.65	

09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.13	6.35	92.44	514.14	213,750.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	36.78	2,025,000.00 *gal*
Application event totals		58.13	6.35	92.44	550.92	

09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	34.33	1,890,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.33	

10/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	34.33	1,890,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.33	

Field 3 - 11/17/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 11/17/2022

Field 3 - 11/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	171.24	20.33	95.43	342.48	171,000.00 *gal*
Application event totals		171.24	20.33	95.43	342.48	

Field 3 - 07/12/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 07/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	82.35	8.99	130.96	728.36	285,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	52.10	2,700,000.00 *gal*
Application event totals		82.35	8.99	130.96	780.47	

07/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	46.89	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	46.89	

Field 3 - 07/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.64	6.30	91.67	509.85	199,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	36.47	1,890,000.00 *gal*
Application event totals		57.64	6.30	91.67	546.33	

09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.64	6.30	91.67	509.85	199,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	36.47	1,890,000.00 *gal*
Application event totals		57.64	6.30	91.67	546.33	

09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.53	5.85	85.12	473.44	185,250.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	33.87	1,755,000.00 *gal*
Application event totals		53.53	5.85	85.12	507.30	

09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	31.26	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.26	

10/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	31.26	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.26	

Field 4 - 11/17/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough

Plant date: 11/17/2022

Field 4 - 11/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	130.47	15.49	72.71	260.94	57,000.00 *gal*
Application event totals		130.47	15.49	72.71	260.94	

Field 4 - 07/12/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 07/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	84.70	9.25	163.59	749.17	128,250.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	53.59	1,215,000.00 *gal*
Application event totals		84.70	9.25	163.59	802.77	

07/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	41.68	945,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.68	

Field 4 - 07/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	56.47	6.17	89.80	499.45	85,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.73	810,000.00 *gal*
Application event totals		56.47	6.17	89.80	535.18	

09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	56.47	6.17	89.80	499.45	85,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.73	810,000.00 *gal*
Application event totals		56.47	6.17	89.80	535.18	

09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	56.47	6.17	89.80	499.45	85,500.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.73	810,000.00 *gal*
Application event totals		56.47	6.17	89.80	535.18	

09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	35.73	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.73	

10/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	35.73	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.73	

Field 5 - 11/17/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough

Plant date: 11/17/2022

Field 5 - 11/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	152.21	18.08	84.83	304.43	57,000.00 *gal*
Application event totals		152.21	18.08	84.83	304.43	

Field 5 - 07/12/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 07/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.84	9.59	139.69	776.92	114,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	55.58	1,080,000.00 *gal*
Application event totals		87.84	9.59	139.69	832.50	

07/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	48.63	945,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.63	

Field 5 - 07/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.86	8.39	122.23	679.80	99,750.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	48.63	945,000.00 *gal*
Application event totals		76.86	8.39	122.23	728.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.90	6.00	87.30	485.57	71,250.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	34.74	675,000.00 *gal*
Application event totals		54.90	6.00	87.30	520.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.90	6.00	87.30	485.57	71,250.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	34.74	675,000.00 *gal*
Application event totals		54.90	6.00	87.30	520.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	34.74	675,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.74	

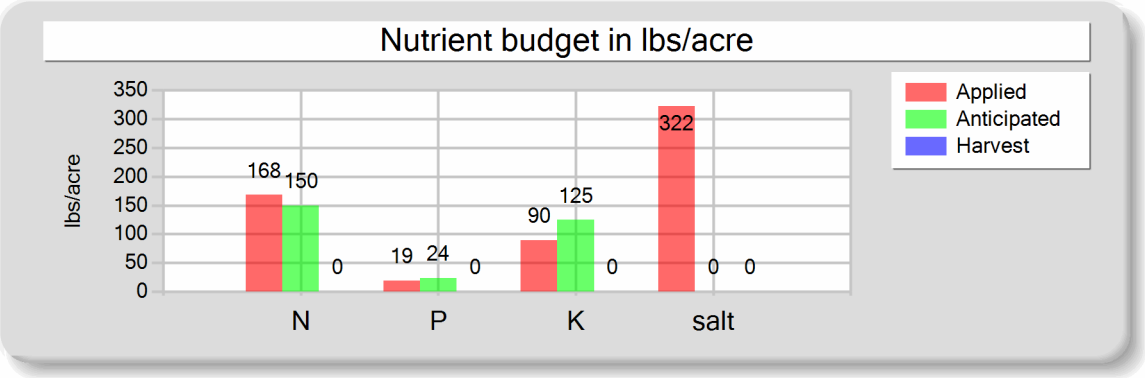
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	27.79	540,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.79	

B. NUTRIENT BUDGET

Field 2 - 11/17/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	161.17	19.14	89.82	322.33
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	168.17	19.14	89.82	322.33
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	168.17	19.14	89.82	322.33
Applied to removed ratio	0.00	0.00	0.00	0.00

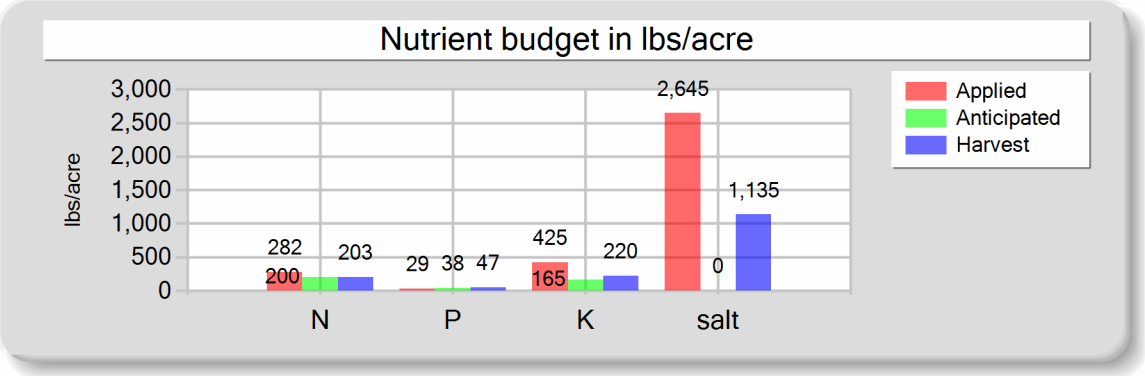
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
171,000.00 *gallons*
6.30 *acre-inches*
0.37 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 2 - 07/12/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 07/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	267.39	29.20	425.22	2,365.03
Fresh water	0.00	0.00	0.00	279.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.39	29.20	425.22	2,644.56
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	203.28	47.29	220.13	1,134.78
Nutrient balance	79.11	-18.09	205.09	1,509.78
Applied to removed ratio	1.39	0.62	1.93	2.33

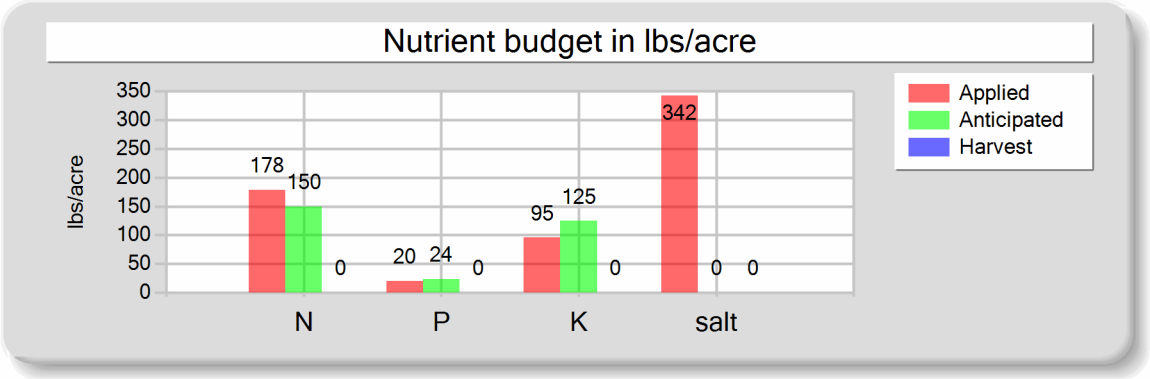
Fresh water applied	
15,390,000.00	*gallons*
566.76	*acre-inches*
33.34	*inches/acre*

Process wastewater applied	
983,250.00	*gallons*
36.21	*acre-inches*
2.13	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/17/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	171.24	20.33	95.43	342.48
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	178.24	20.33	95.43	342.48
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	178.24	20.33	95.43	342.48
Applied to removed ratio	0.00	0.00	0.00	0.00

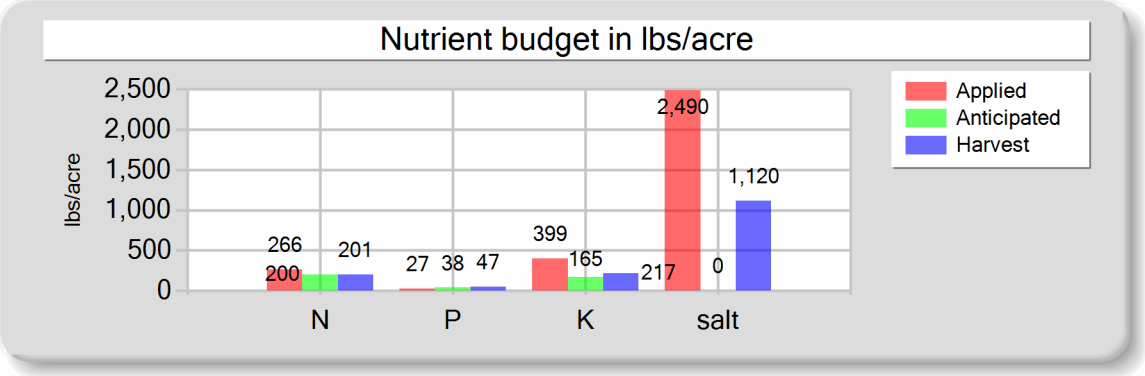
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
171,000.00	*gallons*
6.30	*acre-inches*
0.39	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 07/12/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 07/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	251.17	27.43	399.42	2,221.50
Fresh water	0.00	0.00	0.00	268.34
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	266.17	27.43	399.42	2,489.84
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	200.56	46.65	217.18	1,119.58
Nutrient balance	65.61	-19.22	182.24	1,370.26
Applied to removed ratio	1.33	0.59	1.84	2.22

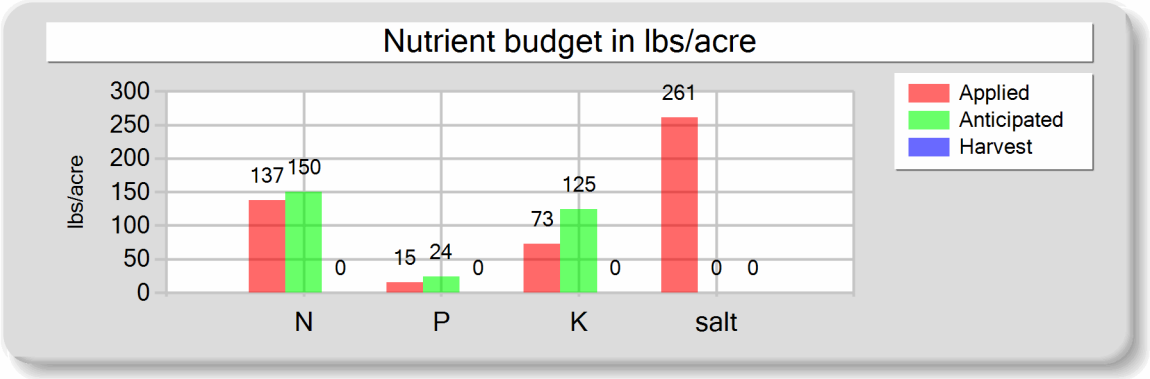
Fresh water applied	
13,905,000.00	*gallons*
512.07	*acre-inches*
32.00	*inches/acre*

Process wastewater applied	
869,250.00	*gallons*
32.01	*acre-inches*
2.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/17/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	130.47	15.49	72.71	260.94
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	137.47	15.49	72.71	260.94
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	137.47	15.49	72.71	260.94
Applied to removed ratio	0.00	0.00	0.00	0.00

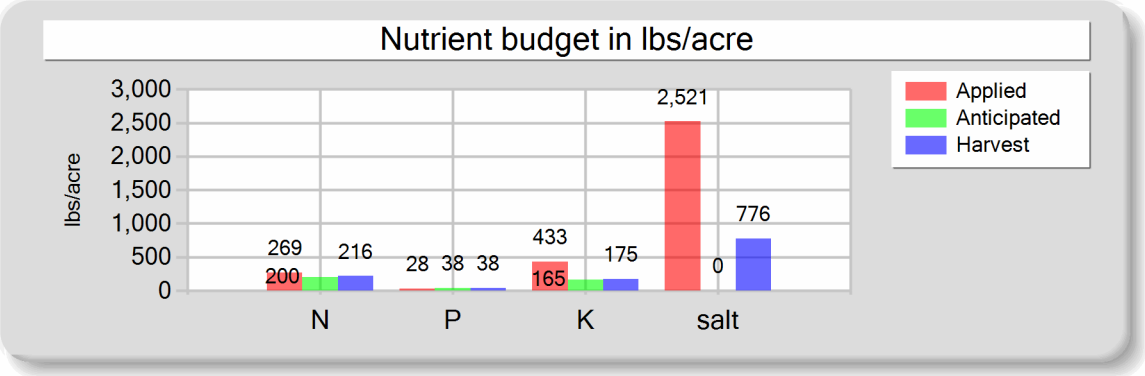
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
57,000.00	*gallons*
2.10	*acre-inches*
0.30	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 07/12/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 07/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	254.11	27.75	432.99	2,247.52
Fresh water	0.00	0.00	0.00	273.92
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	269.11	27.75	432.99	2,521.44
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	216.01	37.67	175.39	775.50
Nutrient balance	53.10	-9.92	257.60	1,745.93
Applied to removed ratio	1.25	0.74	2.47	3.25

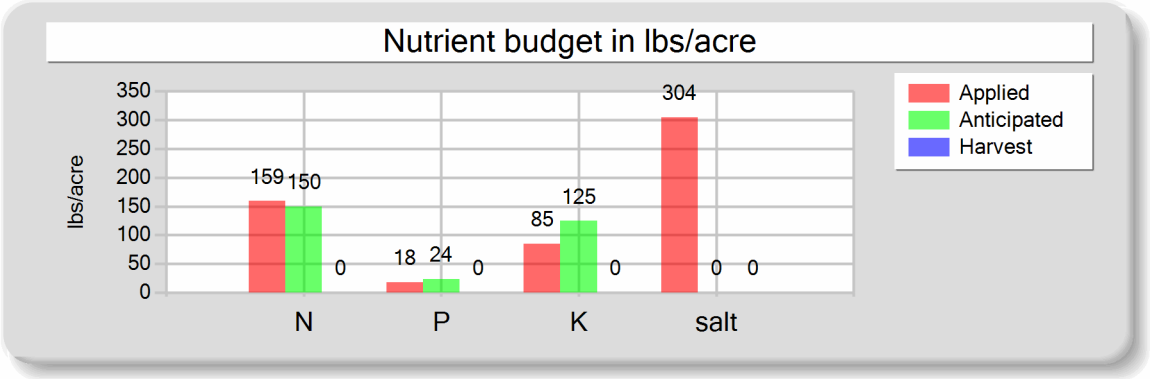
Fresh water applied	
6,210,000.00	*gallons*
228.69	*acre-inches*
32.67	*inches/acre*

Process wastewater applied	
384,750.00	*gallons*
14.17	*acre-inches*
2.02	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 11/17/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	152.21	18.08	84.83	304.43
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	159.21	18.08	84.83	304.43
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	159.21	18.08	84.83	304.43
Applied to removed ratio	0.00	0.00	0.00	0.00

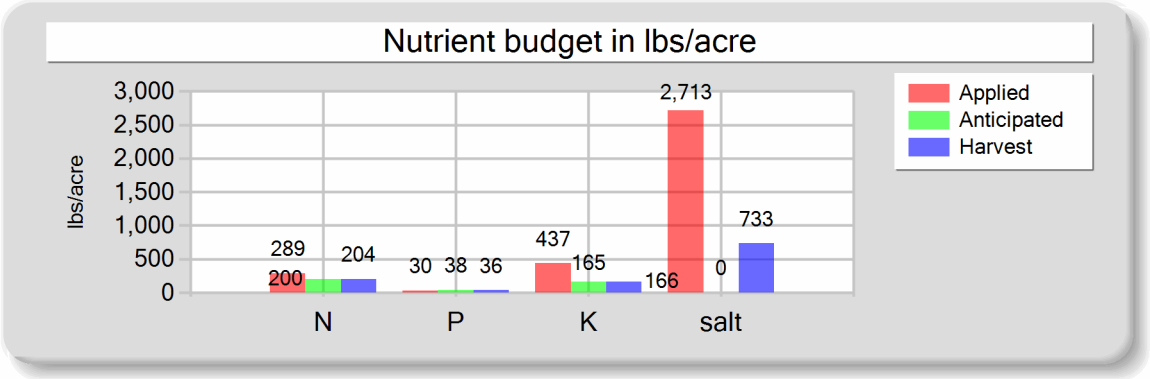
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
57,000.00	*gallons*
2.10	*acre-inches*
0.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 07/12/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 07/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	274.50	29.98	436.52	2,427.87
Fresh water	0.00	0.00	0.00	284.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.50	29.98	436.52	2,712.71
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	204.30	35.63	165.89	733.46
Nutrient balance	85.20	-5.65	270.64	1,979.25
Applied to removed ratio	1.42	0.84	2.63	3.70

Fresh water applied	
5,535,000.00	*gallons*
203.84	*acre-inches*
33.97	*inches/acre*

Process wastewater applied	
356,250.00	*gallons*
13.12	*acre-inches*
2.19	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Manure

Sample and source description: Corral Manure

Sample date: 05/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,200.00	9,040.00	29,100.00							53.40
DL	100.00	100.00	100.00							0.01

Corral Manure

Sample and source description: Corral Manure

Sample date: 10/23/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 31.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	5,470.00	25,900.00							42.60
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ1

Sample and source description: WWQ1

Sample date: 02/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,920.00	526.00	0.00	0.00	228.00	1,070.00								7,540.00	3,840
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 06/02/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,240.00	515.00	0.00	0.00	74.50	1,000.00	163.00	121.00	485.00	4,340.00	0.00	247.00	827.00	8,310.00	4,400
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	554.00	375.00	0.00	0.00	60.50	881.00								8,190.00	4,900
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	473.00	308.00	0.00	0.00	103.00	779.00								7,300.00	4,960
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

MID Lateral 5

MID Lateral 5

MID Lateral 5

Sample description: MID Lateral 5

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								38.10	37
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 2 - 07/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,740.00	870.00	4,050.00		7.30
DL	100.00	100.00	100.00		0.01

Field 3 - 07/12/2023: Corn, silage

Field 3 - 07/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,740.00	870.00	4,050.00		7.30
DL	100.00	100.00	100.00		0.01

Field 4 - 07/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,670.00	640.00	2,980.00		5.40
DL	100.00	100.00	100.00		0.01

Field 5 - 07/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,670.00	640.00	2,980.00		5.40
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

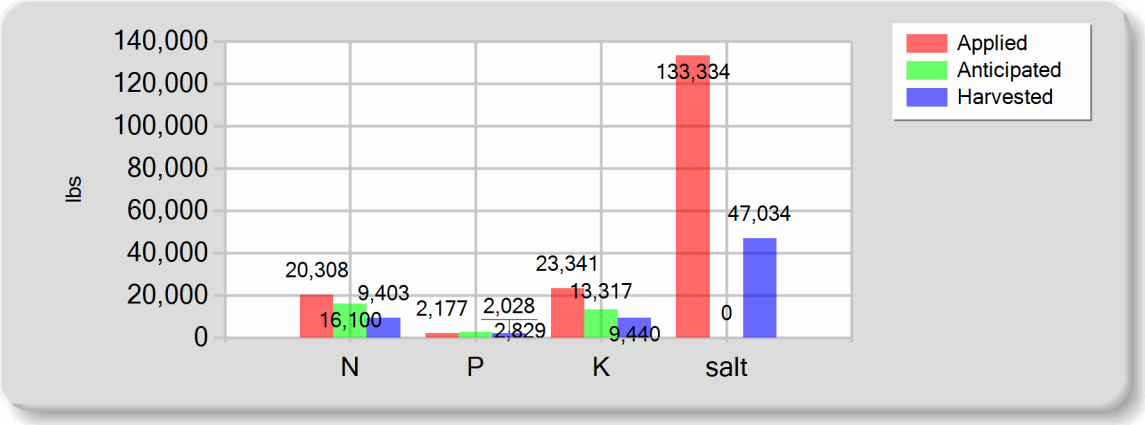
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

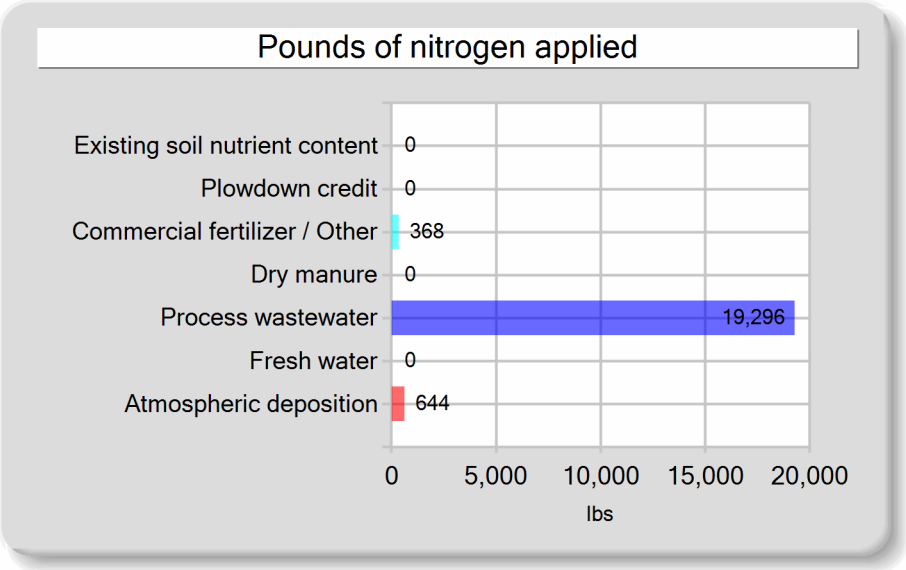
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

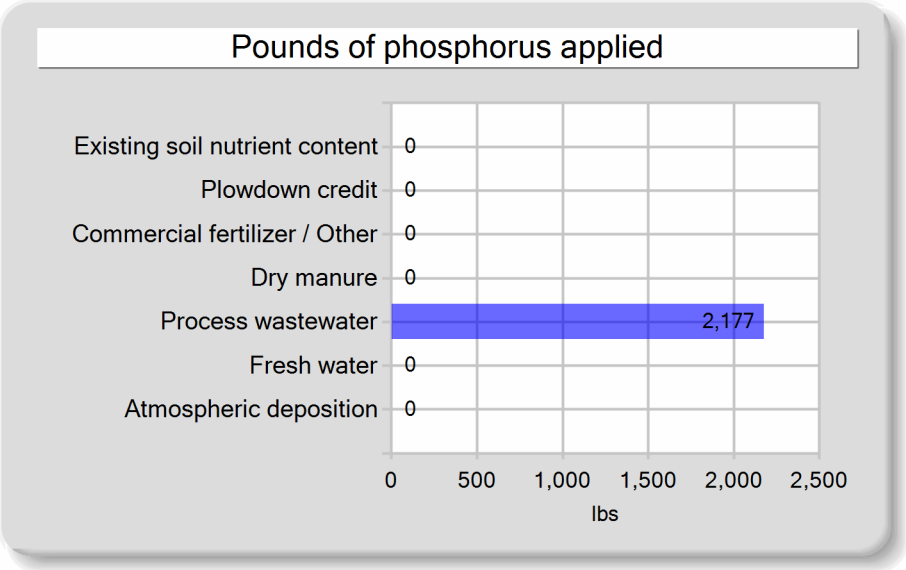
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	368.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	19,296.30	2,177.00	23,341.24	120,661.94
Fresh water	0.00	0.00	0.00	12,671.72
Atmospheric deposition	644.00	0.00	0.00	0.00
Total nutrients applied	20,308.30	2,177.00	23,341.24	133,333.66
Anticipated crop nutrient removal	16,100.00	2,829.00	13,317.00	0.00
Actual crop nutrient removal	9,402.50	2,027.78	9,440.18	47,033.89
Nutrient balance	10,905.80	149.22	13,901.06	86,299.76
Applied to removed ratio	2.16	1.07	2.47	2.83

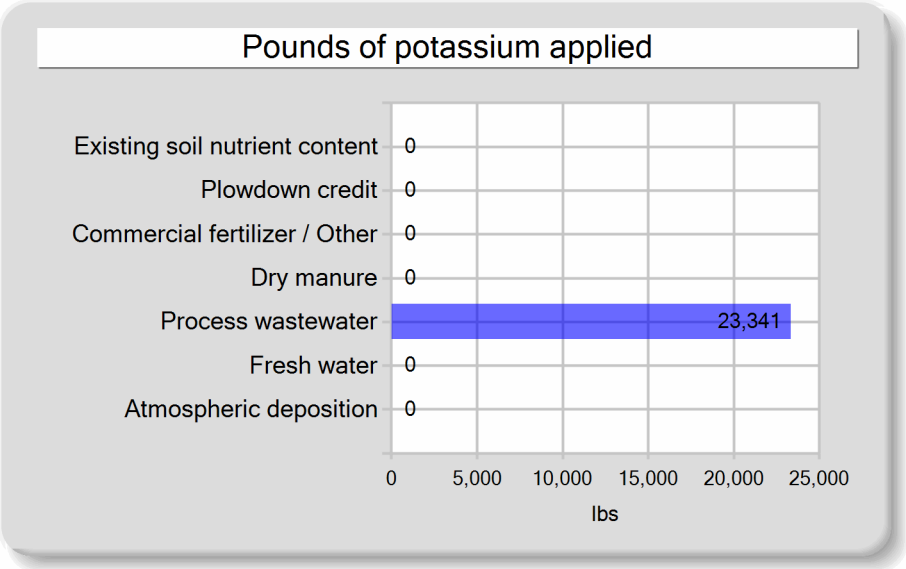
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

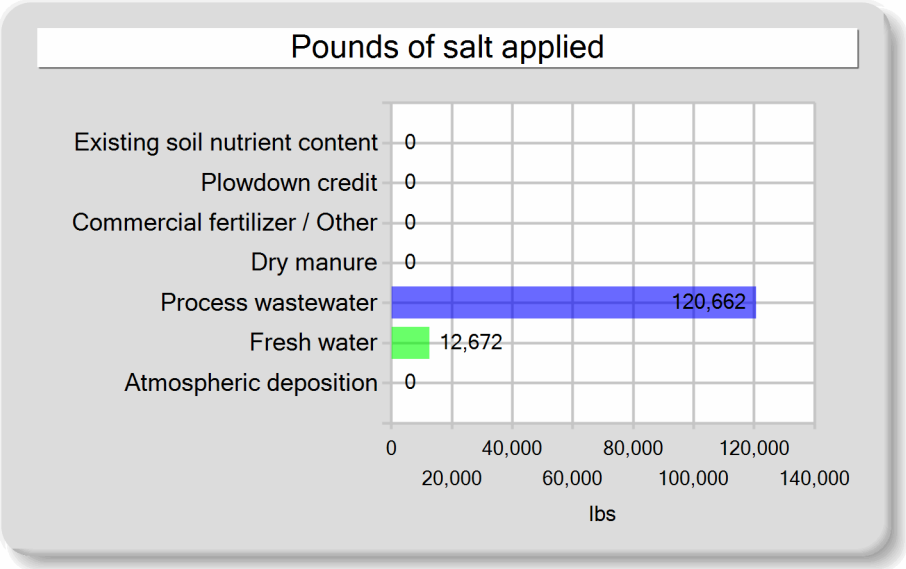


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Due to the 2023 flooding of the Tuolumne River the Oat Silage crop on Fields 2,3,4 & 5 resulted in crop failures.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

John Morris

SIGNATURE OF OWNER OF FACILITY

SIGNATURE OF OPERATOR OF FACILITY

John Morris

PRINT OR TYPE NAME

SAME AS OWNER

PRINT OR TYPE NAME

6/17/24

DATE

DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: John Morris

Name of Dairy Facility: JNM Dairy

Facility Address:

Number and Street	City	County	Zip Code
7742 Paradise RD	Modesto	Stanislaus	95358

Contact Person Name and Phone Number: John Morris (Name) — (209) 404-3909 (Phone Number)

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Morris Dairy Farms

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
3513 Shiloh RD	Modesto	CA	95358

Contact Person: John Morris (Name) — (209) 404-3900 (Phone Number)

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Manuel Morris (Name) — (209) 541-0627 (Phone Number)

Address	City	State	Zip Code
3513 Shilo RD	Modesto	CA	95358

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code

Street and nearest cross street (if no address)	County

Assessor's Parcel Number	Assessor's Parcel Number County
0016-0033-0004-0000	Stanislaus

Last date hauled: 05/17/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,610.00 tons

Manure Solids Content: 51.2 %

Method used to determine amount of manure:

161 Loads @ 10 Tons Per Load

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

John Mains — 6/17/24
Operator Signature — Date

John Mains — 6/17/24
Hauler Signature — Date

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
JNM Dairy
7742 Paradise Rd
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/23/23 3:30 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - House	T3082013A	Electrical Conductivity (EC)	EPA 120.1 and	1.69	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	18.9	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1080	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101			
Electrical Conductivity (EC)	102			3.27
Nitrate Nitrogen (NO3-N)	103	97.4 (T3082012A)	98.0 (T3082012A)	0.634

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
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Compliance Analysis Report

Customer
JNM Dairy
7742 Paradise Rd
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/23/23 3:30 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - Heifers	T3082013B	Electrical Conductivity (EC)	EPA 120.1 and	1.48	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	21.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	945	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101			
Electrical Conductivity (EC)	102			3.27
Nitrate Nitrogen (NO3-N)	103	97.4 (T3082012A)	98.0 (T3082012A)	0.634

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Denele Analytical, Inc.

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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
JNM Dairy
7742 Paradise Rd
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/23/23 3:30 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - Milk Barn	T3082013C	Electrical Conductivity (EC)	EPA 120.1 and	1.62	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	19.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1040	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101			
Electrical Conductivity (EC)	102			3.27
Nitrate Nitrogen (NO3-N)	103	97.4 (T3082012A)	98.0 (T3082012A)	0.634

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.
865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: JNM DAIRY
Mailing address:
(not facility address) MODESTO CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Sean Locke
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	3/23/23	1:00	DW - HOUSE	DOMESTIC	18.8°		X				
2		1:05	DW - HEIFERS		18.6°		X				
3		1:10	DW - MILK BARN		19.0°		X				
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	3-23-23 3:30			T3082013

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)